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RESEARCH

The cost of language: tokenization as a metric of labor

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Abstract
This study analyzes tokenization in Natural Language Processing (NLP) at the intersection of language, technology, and political economy. Recent studies show that subword tokenization in multilingual Large Language Models (LLMs) distributes processing costs unevenly across languages, as non-Latin scripts are disproportionately fragmented by statistical segmentation. Taking this disparity as a point of departure, the study reconceptualizes tokenization beyond a purely linguistic or computational problem. The analysis approaches tokenization through the epistemological problems of translation and situates it within a longer history of socio-technical infrastructures from telegraphy to machine learning that mediate between linguistic abstraction and regimes of social productivity. Rather than treating tokens as neutral technical units, it shows that they function as operational measures of linguistic value beyond meaning, enabling the quantification and automation of linguistic labor. Drawing on Marxist and structuralist accounts of linguistic value, and postcolonial critiques of semiotic standardization, the study frames tokenization as both a technical abstraction and a metric in the social division of labor.

Keywords Tokenization · Multilingual AI models · Translatability · Linguistic value · Political economy of language · Telegraphy

1 Introduction

How much does an automatic translation service or a virtual assistant cost? It might seem like a trivial question, but the answer depends on the language in use. In the age of Artificial Intelligence, language finds itself at the intersection of material production and processes of valorization between labor and market. The fact that languages have a cost, and that this cost discriminates against some languages in favor of others, is a problem that not only concerns today's political economy but also affects the very relationship between language and technology. This dynamic has been in play since at least the introduction of the telegraph, a precursor to digitalization, and the spread of Morse code along colonial routes.

A recent survey on commercial offerings for the use of large language models has highlighted disparities in processing costs for different languages.[1] A team of researchers at the University of Washington (Ahia et al. 2023) conducted a comparative analysis of the costs of OpenAI's language model API on a sample of 22 typologically diverse languages, noting the non-uniformity of prices and, consequently, the specific costs for each language. Researchers at Washington University tell us that "[w]hile LMs like ChatGPT might perform tasks in Telugu, for example, a user in Andhra Pradesh might pay 5× more than an English user in the US for an equivalent use of the model" (Ahia et al. 2023).

This disparity is the result of the LLM's *subword tokenization system*, which generates longer strings of units (tokens) in certain language scripts compared to others. Tokenization in Natural Language Processing refers to the process of segmenting text into atomic units. These units do not necessarily correspond to linguistically distinctive units and are mostly devoid of meaning in a strict sense, as they are the result of a statistical processing of text occurrences. These same occurrences are subsequently used by a trained

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[1] The language models in question, such as the GPT (Generative Pre-trained Transformer) series by the giant OpenAI, now in its fifth generation, are the foundation of popular chatbots like ChatGPT. Companies involved in the development of Artificial Intelligence offer access to language models for individual consumers in the form of a free "basic" or limited version, and through paid subscriptions for the latest generation or "plus" versions, with unlimited use, following the well-established "try and buy" formula in the digital economy. To use

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language model to predict the next token, that is, the most plausible unit following a given sequence of signs, thereby generating new text. This is how ChatGPT "speaks" to us. However, what constitutes a token is not a universal standard and is still a subject of ongoing research and debate (Mielke et al. 2021). Various design choices can affect the number of tokens in a language's writing system, leading to increased fragmentation. Over-tokenization can result from a scarcity of related training data, as in the case of "low-resource languages" (Caffoni 2025), because fewer recurring patterns are available from which stable subword units can be learned. It can also result from the specific properties of individual writing systems or the way they are represented in the Unicode encoding system, the universal character classification standard for electronic text processing.

The API of LLMs charges users a fixed amount for a certain quantity of input and generated tokens.[2] The subword tokenization system of the ChatGPT API, for example, creates disparities in costs between languages: Latin scripts typically encounter an average lower fragmentation rate compared to that of other languages. For instance, Telugu, which has about 96 million speakers in Central–Southern India, requires five times more tokens to convey the same "information" compared to English, thereby exponentially increasing the specific costs per language (Ahia et al. 2023). The ratio with English is 1 to 15 for Shan with approximately 4.7 million native speakers in Eastern Myanmar, 1 to 3 for Arabic, and also 2.6 for Bulgarian and 1.6 for Italian (Petrov et al. 2024).

Should OpenAI be more transparent with its customers regarding the economic and performance disadvantages that occur in the Telugu language compared to English? This appears to be the conclusion reached by researchers at Washington (Ahia et al. 2023).[3] Another research group at the University of Oxford has proposed the goal of achieving "tokenization parity" among languages (Petrov et al. 2024), a form of equal opportunity policy for computational linguistics, where tokenizers produce similar encoded lengths across languages, redistributing some of the privileges currently held by English to other languages.

Instead of asking how the restructuring of the commercial offering could enable a fairer distribution of resources among language communities, or whether tokens should serve as a new form of "general equivalent" among languages, I shift the question to tokenization itself. As a form of abstraction and a metric applied to language in the market, tokenization should be discussed within the political economy of the sign. In parallel, in this article, I propose to reconsider the form of abstraction introduced by subword tokenization through the epistemological problems of translation (Liu 1999a, b), and I argue that a translational logic continues to structure tokenization even in ostensibly monolingual processing. These two analytical frameworks, translation and political economy, are developed as complementary perspectives whose relation is elaborated in the following sections.

Read through the lens of translation, the origins of subword tokenization appear less as a neutral optimization than as a response to a specific multilingual constraint. Subword tokenization first emerged as a necessary solution to the multilingual challenges posed by Neural Machine Translation (Sennrich et al. 2016). In 2016, Sennrich and colleagues recognized that translation is an "open vocabulary problem." While most models of the time were limited to 30,000–50,000 words, none of them was capable of tackling the process of word formation such as compounding and agglutination. "[T]ranslation of some words is transparent in that they are translatable by a competent translator even if they are novel to him or her, based on a translation of known subword units such as morphemes or phonemes" (Sennrich et al. 2016), the case with name entities, cognates, and loanwords, and morphologically complex words. Therefore, the researchers conclude, translation models require mechanisms that go below the word level.

Footnote 1 (continued)

the resources offered by this technology not as end consumers, but as an integrated tool in the development of additional applications, we instead need to use an Application Programming Interface (API), which allows communication between our software and OpenAI's proprietary language model.

[2] The pricing list for GPT is published, although it does not account for the differences arising from text fragmentation in different languages, accessed January 7, 2025. https://openai.com/api/pricing/

[3] Interestingly, even character-level tokenization is not a straightforward fix for this problem, as character usage varies significantly across scripts, with some languages being more concise than others (Petrov et al. 2023; Ahia et al. 2023). Toolkits, such as spaCy historically developed around rule-based and pipeline-oriented NLP, primarily implement surface-level word tokenization (segmenting text into whitespace- and rule-delimited word units), rather than the statistically learned subword segmentation schemes used in most large language models (Honnibal 2015). Word-level segmentation has been shown to introduce open-vocabulary Querychallenges for statistical language modeing (Mielke et al. 2021), even as such toolkits increasingly interface with neural architectures. Further complications arise when open-source models are taken into consideration, as many are distributed under ML-specific licenses that impose restrictions on downstream use, including limitations on commercial deployment, such as OpenRAIL or LLaMA licenses (Stalnaker et al. 2025). Computation under more permissive licenses, such as Apache 2.0, still burdens developers with the cost of hardware and energy consumption required to run the model. These constraints illustrate how engineering solutions, while valuable, remain embedded within broader infrastructural and political economies. In this study, I do not pursue technical alternatives; rather, I propose reading tokenization as a form of segmentation that structures both language and labor. While not a technical fix, this offers an alternative framework for further dialogue.

Even within a monolingual processing context, both morphological variation and the presence of rare words can be understood as consequences of the internal and external diversity of language. Internally, factors such as dialectal, stylistic, and domain-specific variation introduce low-frequency lexical forms and morphological complexity. Externally, translation contexts reveal structural and lexical divergences across languages, often resulting in sparsity or unpredictability at the word level. Subword tokenization emerges as a response to these pressures not only to manage vocabulary size or improve efficiency, but also to accommodate the variability inherent in language systems operating across different linguistic and communicative environments.

2 Alphabetic labor time

To grasp the impact of abstraction in translation and its role in the political economy of language, it would be useful to recall that disparities in "the cost of language" didn't appear anew with LLMs. When telegraphy was first introduced in China during the late Qing Dynasty with the installation of the first line between Shanghai and Hong Kong in 1871, carried out by two foreign companies, the Great Northern Telegraph Company of Denmark and the Eastern Extension A&C Telegraph Company of the United Kingdom, we first encounter what Thomas Mullaney has named a problem of "semiotic sovereignty" which continues to haunt information technologies in China to this day (Mullaney 2014, 2018). Even after the infrastructures of telegraphy were brought under Chinese ownership, the problem of the Chinese telegraphy code remained unresolved. "The most basic functionality of Chinese telegraphy depended upon the creation of a novel information-semiotic system, i.e., a Chinese telegraph code" (Mullaney 2014, 155). Although Morse code was devised specifically for the phonetic structure of the English alphabet, its binary character later came to be interpreted as language-invariant. This retrospective framing renders invisible the hegemony of a particular phonetic system over the diversity of global scripts. To bypass the limitations of Morse code, in The Chinese Telegraph Code of 1871, approximately 6800 characters were first assigned in a four-digit Arabic numeral sequence from 0001 to 9999; the numeral sequence was then used to transmit a message using the standard Morse signal. The double encoding of the Chinese scripts rendered telegraphy expensive, as not only were Arabic numerals the lengthiest sequences to be transmitted in Morse, and therefore more costly, but the time required to encode and decode the binary signals first in numerical sequences and then into characters also increased labor costs across the network.

The arbitrary assignment of numerical codes to characters eliminated any significant connection between the characters and their corresponding codes, transforming the language into an alphanumeric system suited to Western standards of communication and translation (Tsu 2022, 108). The crisis faced by the cultural prestige once associated with the Chinese script, amid the practical challenges brought on by telegraphy, can be interpreted as one of many instances of the imposition of "material and technological preconditions for importing and transforming foreign knowledge" (Tsu 2014, 115). The labor required to master the writing system had become a liability in the face of the pragmatic need for quick access and efficiency, which permeated the ideal of universal communication at the turn of the century. It was at this juncture that the old woes expressed by foreign missionaries, who had painstakingly learned the difficult language since the time of Matteo Ricci, were internalized by Chinese language reformers in response to their own sense of urgency. The necessity of adapting to Western technological compatibility sparked an extended period of experimentation, both linguistic and technological, beginning in the 1880s and continuing for more than 130 years into the twenty-first century (Tsu 2014, 116).

In a recent volume titled *Codes of Modernity* (Kuzuoğlu 2023), historian Uluğ Kuzuoğlu hypothesizes that, in the incommensurability of Chinese writing in relation to Morse code, an inefficiency in labor time comes into view. This inefficiency, he argues, is determined by the imposition of a new order linked to a specifically Western historical experience. That order was shaped by a political economy of information structured around alphabetic technologies such as the telegraph, and by a discipline of physical and mental labor that identified Chinese writing as an obstacle to the speed of information transmission. "The historical emergence of 'phonocentrism,' in other words, must be traced to the turn-of-the-century clerical and mental labor regime that was grounded in the political economy of the global information age" (Kuzuoğlu 2023, 12). In the mid-nineteenth century, the new discipline of labor which emerged with the industrialization of China's economy, along with new practices of bureaucracy and telegraphic communication, constituted the historical conditions that turned "efficiency" into a normative marker of superiority, and exerted increasing pressure on the Chinese language toward what Kuzuoğlu termed "alphabetic labor time" (Kuzuoğlu 2023, 12).

The intrinsic discriminations associated with alphabetic labor time and the relationship between capital and information have a long history. Although they have particularly affected Chinese characters, at least until the introduction of predictive text functions, known as "autocomplete," in the digital age, techno-linguistic inequalities have impacted many writing systems other than the Latin alphabet. One need is only to consider the use of *jamo* (자모) in Korean, a compound term combining *ja-eum* (자음, consonants) and *mo-eum* (모음, vowels). These orthographic units, which

also carry phonemic value in the Korean alphabet (*Hangul*, 한글) are not arranged in a simple linear sequence like letters in the Latin alphabet; rather, they are combined both horizontally and vertically to form syllable blocks.[4] The ligatures and contextual variations in Arabic script, where letter forms shift according to positional context within words, were ill-suited to Morse code, which was conceived with Latin non-cursive characters in mind. Diacritical marks and the non-linear arrangement of vowels in the Thai alphabet, for example, necessitated transliteration into the Latin alphabet, introducing notable issues with accuracy and leading to a significant reduction in tonal richness and a simplification of transmitted messages. Although to a lesser extent, the accentuation systems in some European languages were similarly viewed as marginal deviations from the norm. French, for instance, was perceived as "English with accents," and it was only in 1903, over sixty years after the advent of electric telegraphy that accented letters were formally incorporated into the "standard" semiotic repertoire (Mullaney 2014, 107).

Matteo Pasquinelli identifies a key role of telegraphy in defining information as a metric of mental labor: "A quintessentially social technology, the telegraph exerted a lasting influence on the disciplines of the mind and the automation of mental labor and communication. Recognizing the telegraph as a *mediating machine* may help us illuminate a historical relation between labor automation and information theory." (Pasquinelli 2022). Telegraphy, therefore, was as much an "epistemic mediator" as it was a "social mediator" playing the role of controlling social productivity. Pasquinelli refers to the distinction between "valorizing information" and "control information" put forward by Romano Alquati in his research on labor composition at the Olivetti computer factory in Ivrea in the 1960s (Alquati 1963). While information emerges from all micro decisions that workers make during the production process, it simultaneously recomposes those same decisions to reorganize the division of labor. In this sense, the quantification of human labor, in the form of a metric system capable of abstracting micro decisions, namely *information*, goes hand in hand with "a *political translation* with deep social implications" (Pasquinelli 2022). But what type of translation are we dealing with?

[4] It should be noted that communication infrastructures have not only been subject to Western colonial interests in Asia, but have also played a key role in competition among neighboring countries. For example, Korea's telegraph lines were used by neighboring rivals Japan and China as a means to seize power and influence over the peninsula (Jin 2017).

3 Automating linguistic value

In *Tokens of Exchange*: *The Problem of Translation in Global Circulations*, Liu (1999a) recontextualizes translatability as a historical event that, in the nineteenth century, inhabited the same order of universalist aspirations as the invention of the metric system, the gold standard, the modern postal service, international law, and the telegraph. In this sense, the designation of token, both in its linguistic meaning as *sign* and its economic meaning as *proof* of value, does not differentiate between symbolic exchange and material circulation, insofar as both signs and objects participate in the logic of equivalence, or non-equivalence, established by exchange value. Liu writes, "I take translation as a primary agent of token making in its capacity to enable exchange, producing and circulating meaning as value among languages and markets" (Liu 1999a, 4). Upholding a postcolonial perspective concerned with exposing the inequalities inherent in the supposed "equivalence" of linguistic exchange in translation, Liu turns to Marx, finding support in the notion of labor time and its objectification in the analysis of commodity exchange. In Marx's labor theory of value (Marx 1976), the quantification of socially necessary labor time is an "average" which does not correspond to the quantity of actual labor required. However, for the circulation of commodities to occur, trade must be conceived as an exchange of equivalents, albeit what is actually in place is an unequal exchange of quantities of labor time. Liu concludes: "Exchange-value is to political economy what simile, metaphor, or synecdoche is to the linguistic realm of signification, as both involve the making of equivalents out of non-equivalents through a process of abstraction or translation" (Liu 1999b, 24). The study of communication is central in Marx, hence his interest in significations and substitutions (i.e., abstract labor for concrete value, exchange value for use value, etc.). It is no coincidence that, in *Capital* Volume I, Marx (1976, 143) uses the term "language of commodities" to identify the property of exchange and communicability in the value equation, which alone can explain how the labor embodied in the coat is equivalent to that in the linen. Interestingly, Marx highlights that the equation between value and different quantities of human labor is not self-explanatory. On the contrary, interpreting the exchange or "communication" established between commodities requires a form of code-breaking akin to translating an unknown language. He writes:

> Value, therefore, does not have its description branded on its forehead; it rather transforms every product of labor into a social hieroglyphic. Later on, men try to decipher the hieroglyphic, to get behind the secret of their own social product: for the characteristic which

> objects of utility have of being values is as much men's social product as is their language (Marx 1976, 167).

Ten years before the publication of *Capital*, Marx had already noted in the *Grundrisse* notebooks that the analogy between money and language can only be understood as one mediated through translation:

> To compare money with language is not less erroneous. Language does not transform ideas, so that the peculiarity of ideas is dissolved and their social character runs alongside them as a separate entity, like prices alongside commodities. Ideas do not exist separately from language. Ideas which have first to be translated out of their mother tongue into a foreign language in order to circulate, in order to become exchangeable, offer a somewhat better analogy; but the analogy then lies not in language, but in the foreignness of language (Marx(Marx 1993), 162–163).

Marx points to translation as the process of abstraction that makes it possible for commodities to talk to each other. Yet, exchange value is estranged from human labor, taking the form of a secretive language that requires deciphering. The act of translation, rather than being understood in terms of labor, is intrinsic to the *money form*, which obscures the true social character of the relationships between producers. Thus, according to Marx, while *reading* and *writing* may be akin to material production in accordance with the laws imposed by nature, capital emerges as the sole translator, and only commodities speak in translation, proclaiming: "[W]e relate to each other merely as exchange-value" (Marx 1976, 177).

Even Ferdinand de Saussure, in the *Course in General Linguistics*, stated that "language is a system of pure values" (Saussure 1983, 80). This perspective allows for the possibility, as Barthes noted, "to de-psychologize linguistics and to bring it closer to economics" (Barthes 1968, 54).

Saussure, like Marx, understood the mutual implication of economic exchange and linguistic exchange: "The reason is that, as in the study of political economy, one is dealing with the notion of *value*. In both cases, we have *a system of equivalence between things belonging to different orders*. In one case, work and wages; in the other case, signification and signal." (Saussure 1983, 79). But whereas for Marx exchange value can be understood and analyzed in terms of abstract labor and use value, for Saussure, linguistic value does not correspond entirely to signification since the arbitrary nature of the sign is only guaranteed by the interdependency of each term in a language. He reaches the problem of value in language to solve another problem, namely that of "poorly defined" linguistic units (Saussure 1983, 109), as these are on the level of speech sounds, words, compounds, sentences or entire languages:

> A language thus has this curious and striking feature. It has no immediately perceptible entities. And yet one cannot doubt that they exist, or that the interplay of these units is what constitutes linguistic structure. That is undoubtedly a characteristic which distinguishes languages from all other semiological institutions (Saussure 1983, 105).

Saussure argues that value is the form which articulates and fixes the unit relation between the "amorphous masses" of thought and sound (Saussure 1983, 111). But despite value being an element of signification, it does not coincide with the latter: "On the one hand, the concept appears to be just the counterpart of a sound pattern, as one constituent part of a linguistic sign. On the other hand, this linguistic sign itself, as the link uniting the two constituent elements, likewise has counterparts. These are the other signs in the language." (Saussure 1983, 113). To illustrate how both features contribute to the definition of linguistic value, Saussure indeed uses the analogy of a coin:

> To determine the value of a five-franc coin, for instance, what must be known is: (1) that the coin can be exchanged for a certain quantity of something different, e.g., bread, and (2) that its value can be compared with another value in the same system, e.g., that of a one-franc coin, or of a coin belonging to another system (e.g., a dollar). Similarly, a word can be substituted for something dissimilar: an idea. At the same time, it can be compared to something of like nature: another word. Its value is therefore not determined merely by that concept or meaning for which it is a token. It must also be assessed against comparable values, by contrast with other words. The content of a word is determined in the final analysis not by what it contains but by what exists outside it. As an element in a system, the word has not only a meaning but also, above all, a value. And that is something quite different (Saussure 1983, 113).

For Saussure, linguistic value in its arbitrariness is essentially a social fact, insofar as it necessitates a community of speakers with their usage and general acceptance. In a linguistic community, words expressing related ideas limit each other reciprocally, as in the case of synonyms, or conversely, some words are enriched by their context, which determines their value. Saussure indirectly highlights the impossibility of absolute equivalence between languages. Translation cannot achieve perfect equivalence, as linguistic value is determined not only by the correspondence between signs and concepts (signification) but also by the specific relationships a sign holds with other values within the same language. "There are languages,

for example, in which it is impossible to say the equivalent of *s'asseoir au soleil* ('to sit in the sun')" (Saussure 1983, 114).

Without engaging in a detailed analysis of the rules that govern the reciprocity of meaning in languages other than French or English (e.g., the different value accorded to "sheep" and "*mutton*"), Saussure approaches translation as a relationship between variable values. It is value which determines the internal structure of a language, and at the same time, it is value that defines the arbitrary boundary that distinguishes and separates one language from another. The same is true for the material aspect of language, where sounds are important only since they define phonetic contrasts and non-coincidences with other sounds. In the same way, "it is not the metal present in a coin to determine its value. A crown piece nominally worth five francs contains only half that sum in silver. Its value varies somewhat according to the effigy it bears. It is worth rather more or rather less on different sides of a political frontier" (Saussure 1983, 117). Yet Saussure failed to recognize the centrality of translation in the cause of defining both meaning and value; a lacuna which will be directly addressed later on by Roman Jakobson's theory of translation (Jakobson 1959).

What Saussure introduced was not entirely novel. While he uniquely framed the relationship between linguistics and economics at a symbolic level (the relationship of values) rather than a pragmatic one (how language functions), this privileged connection has deep historical roots. As Maria Grazia Turri reminds us, until the mid-nineteenth century, philosophers and economists belonged to the same field of study: moral philosophy (Turri 2015).[5]

There has been much debate about the resurgence of structuralist paradigms of language in light of the recent success of LLMs (Weatherby 2024; Gastaldi 2020). These interpretations do not reduce LLMs to Saussurean structural dualities; rather, they reintroduce the problem of language as a system of values, in contrast to cognitive approaches that treat language as the primary route to modeling intelligence. Take for instance, the statistical values assigned to tokens as a unit measure. In this sense, the *structuralist hypothesis,* namely that meaning is the effect of structure (Gastaldi 2020) when applied to LLMs leads us to explore the automation of value formation.

On the other hand, the ubiquitous application of LLMs has raised epistemological concerns on the use of structural linguistics as a "meta-science" which assumes, inaccurately, that LLMs can serve as general pattern machines (Offert et al. 2026). The criticism of the language paradigm leads to the question of the specific knowledge that transformers afford, beyond the "hasty projection", I recast Fredric Jameson's expression that comes with the use of metaphors to describe the operation of transformer architectures. Nevertheless, highlighting the transformer "non-linguistic" treatment of language (Offert et al. 2026) is possible only by assuming an *image* of language which is confined to the realm of signification. Both approaches (Gastaldi 2020 and Offert et al. 2026) seem to emphasize the centrality of tokenization in the current stage of research in NLP, albeit with diametrically opposed conclusions. It is no coincidence that even Saussure placed the definition of linguistic units at the heart of the issue of value.

Rather than asking whether transformers are *really* models of language, we could investigate what transformers reveal to us about language itself, e.g., the possibility to isolate discrete units (tokens) without any further reference to signification. As Juan Luis Gastaldi has effectively highlighted, readdressing our inquiry toward natural languages involves substituting the question "Why can computers understand natural language?" with "What must natural language be for those mechanisms to succeed in the way they do?" (Gastaldi 2020). This does not imply ceasing to question the epistemological premises set in motion by the machine. On the contrary, once we set aside the restriction of semantics, it becomes possible to formulate the query: What is the specific unit that the token measures once we move away from the domain of meaning? What are the social relations prescribed by the automation of linguistic value?

[5] One need only consider Adam Smith, who regarded linguistic exchange as the first form of commerce. Literature has also played its part; most notably, we recall the famous episode in Goethe where Mephistopheles attempts to convince Faust of the dissociation between word and concept, an idea that influenced Marx's own reflections on linguistic and monetary alienation. The relationship between capital and language is also traceable, after 1968, in Jean Baudrillard's political economy of the sign (Baudrillard 1972), in the "semantic materialism" of the *Tel Quel* group (Kohlbry 2022), in streams of literary criticism from the 1980s such as Marc Shell and Jean-Joseph Goux's *New Economic Criticism* (Shell 1982; Woodmansee and Osteen 1999), in Italian *post-operaismo* (post-workerist thought) from figures like Paolo Virno and Christian Marazzi, and in the *Langage et Travail* group in France during the 1990s. More critical of analogies, isomorphisms, and homologies between different levels of reality, however, was Fredric Jameson, who warned of the risks and limits of "hasty projections" (Jameson 1972) that have long inhabited the theoretical articulation of the universal equivalent: attempts to extract linguistic structures from economic realities risk reducing them to mere conceptual abstractions. "In practice, indeed, it turns out that it is much easier to extract linguistic structures from cultural objects which are already of a linguistic character, than from the economic realm itself: Such homologies often prove to be little more than hasty projections of the former upon the latter, so that the subsequent 'identity' between the two thus does not really come as much of a surprise. Even where this is not the case, there remains the danger that the identity holds good, not for the concrete realities themselves, but merely for the conceptual abstractions that have been derived from them" (Jameson 1972, 214).

4 Language and division of labor

The philosopher of language Ferruccio Rossi-Landi reproached Saussure for reducing linguistic value to exchange value, thereby excluding the inquiry into the social production of language and the form of social relation within which the linguistic value is produced.

> Saussure himself, though he deals with them amply, does not seem to have a theory of linguistic work, which alone could give a foundation to his theory of linguistic value (Rossi-Landi 1983, 50).
> To go back for a moment to Saussure, or perhaps better to his interpreters, it is indeed singular that in language [*langage*] they saw only the mere sum or co-presence of *langue* and *parole*, and then declared this sum to be "unknowable." Similarly, it is highly revealing that *parole* was taken to be necessarily individual, whereas collective speech was got rid of by inventing the ambiguous, amorphous notion of a "speaking mass." Such a schema has a marked ideological imprint, the nucleus of which is constituted by a refusal to recognize the founding power of work and thus also the explicative and revolutionary power of the notion of work (Rossi-Landi 1983, 151–152).

Despite conceiving the system of linguistic value as socially produced, Saussure confines the acts of speaking (*parole*) to the domain of the individual. Rossi-Landi's intention was to overcome the limitations of theories that described communication at the level of the "linguistic market," oriented toward the "pure economy" of Marginalism advanced by the Lausanne School (Léon Walras, Vilfredo Pareto) (Petrilli 2018). Taking into consideration Marx's critique of political economy, Rossi-Landi argues that the study of language must shift from analyses of the *linguistic market* to that of *linguistic labor*. According to Rossi-Landi, at the level of labor, it is possible to identify a "homology of production" between material artifacts, such as wooden planks, shoes, or cars, and linguistic artifacts, such as words, sentences, or discourses (Rossi-Landi 1974, 1852). This *homology* can, in turn, be distinguished from *analogy* and *isomorphism* because it concerns structural correspondences rather than a posteriori similarities, which Jameson criticizes (1972). These correspondences arise both at the level of historical formation (phylogenetic) and of internal developmental process (ontogenetic). The one between linguistic production and material production is a constitutive relation. Rossi-Landi writes, "Man has never produced linguistic artifacts without producing material artifacts at the same time" (Rossi-Landi 1974, 1852).

The hypothesis that language plays a role in the division of labor was already explored in the 1930s by U.S. linguists such as Leonard Bloomfield, who argued that "the division of labor, and, with it, the whole working of human society, is due to language" (Bloomfield 1961, 24). Yet Rossi-Landi critiques Bloomfield's prioritization of linguistic labor over non-linguistic labor, instead emphasizing the complementarity that characterizes the homology of production. As he states: "language in its turn, and with it the whole working of human society, is due to the division of labor" (Rossi-Landi 1974, 1853 note 23). This suggests that the homology between linguistic and material production is realized not only on the logico-structural plane but also through a historical-genetic method.

Rossi-Landi finds confirmation of his hypothesis in the research on the origins of language conducted by the Vietnamese philosopher Trần Đức Thảo.[6] Rossi-Landi takes an interest in Thảo's research with whom he had been in correspondence in the early 1970s to support his argument about the homology between language and labor. Still in the late 1960s, Rossi-Landi did not clearly define the origin of this homology, at the risk of working theoretically with an abstract conception of labor as akin to any human activity (D'Alonzo 2017). Thảo investigated the symbolism of language and its imaginary transposition of productive activities that is how he writes "the animal cry becomes human language by being articulated within the structure of productive labor" (Thảo 1986, 171). Similarly, when a child learns to speak, the utterance of a vocal sign appears as a sketch of usage that ideally mediates the desired effect, as in the case of the sound made by an infant when a parent interrupts the game of repeatedly turning a lamp on and off. The child's vocal emissions, before functioning as words that designate conceptual structures, refer to concrete means of acting upon objects and thus appear as symbolic forms of instrumental efficacy. This original experience can also be understood in

[6] Trần Đức Thảo sought to reconcile phenomenology and dialectical materialism in his study of the origins of language and consciousness. *Phénoménologie et matérialisme dialectique* was published in France in 1951, the same year Thảo returned to Vietnam to join the struggle for national liberation. In his critique of Saussure's theory of the arbitrariness of the sign and value, Thảo argued that arbitrary systems of signs ultimately rest on fundamental natural signs – such as gestures, vocalizations, and physiognomic expressions. These signs, he maintained, possess a distinctly corporeal nature and are embedded in practical, collective life. According to this view, every discourse about the world involves the execution of a more or less defined "indicative gesture" that underscores the subject's interaction with their surrounding environment (Thảo 1984; D'Alonzo 2020). In the 1970s, Silvia Federici interpreted Thảo's work as an effort to rescue linguistic philosophy from modern imperialism and the bourgeois character inherent in theories of the arbitrariness of the sign. As Federici noted: "[I]f we cannot grasp reality as such, every system (political, epistemological, and otherwise) is as arbitrary as the next, thus capitalism is at least as good as communism, and, to the extent that it is the existing one, it could never be replaced on rational ground" (Federici 1970, note 18).

everyday adult life, as with the cry "heave ho!" Thảo notes that this exclamation arises from the objective structure of a collective pulling effort and is followed immediately by the resumption of the action (Thảo 1986, 237 note 3).

> [T]he utterance of the verb results from rhythmic halts in the collective use of the tool, as a *sketched out movement of production*, ideally ending in the productive operation itself, and experienced as a *signified intention*. The ideal work of the concept is but the very movement of real work being interrupted for an instant by virtue of its objective structure and being continued on the symbolic plane by the use of the word (Thảo 1986, 170).

It should be noted that for Thảo language plays a homological function to that of labor, because it results from the internalization of behaviors related to the collective production and use of tools. In other words, labor has a genetic primacy over language. By contrast, for Rossi-Landi, the two activities share the same structure which characterizes human nature (D'Alonzo 2017).

5 Is linguistic labor universal?

Both Thảo and Rossi-Landi, despite certain differences, seem to inscribe language into a teleology of its productive functions. Thảo, who opposed Jakobson's idea that the meaning of a sign is given in its translation into another sign (Thảo 1984, 33), reaffirmed the primacy of propositionality and the denotative function of language over the interpretation of a sign carried over by other signs. Rossi-Landi, on the other hand, draws on the homology between language and labor to define "common speech," in opposition to Saussure's *parole*. This common speech refers to the similarity of functions performed among different languages, rooted in the similarity of the relation of production and the social constitution of different groups of people (Rossi-Landi 1983, 40). According to Rossi-Landi, it is this common speech that ensures the translatability of languages, while the untranslatable rooted in the differences between specific languages occupies a somewhat minor position.

> The very presence of thousands of languages, each one different from the others, yet all fulfilling the same functions, those of satisfying the basically similar needs for expression and communication, is certainly not the product of any mythical unity of origin of all languages and even less of a natural-law unity of the human species. The similarity of the functions fulfilled by the various languages is derived from the fact that in the process of language development the general forms of social formation, that is, the basic work and production relationships that separate any human society from any pre-human (only animal) society are necessarily represented (Rossi-Landi 1983, 41).

The limit of the translatable is thus inscribed in the materiality of the relations of production and the functions of social life common to the given linguistic communities in question. For Rossi-Landi, this is indeed a "material translation" that takes place on the plane of articulations of both material production and linguistic production (Rossi-Landi 1983, 144; Rossi-Landi 1974, 1900). In this sense, translation is only one of the various instances in which linguistic labor explicitly and self-evidently manifests itself, as in the case of learning a new language, professional translation work, when speaking or writing becomes an end in itself, as in the construction of the poetic message, or in the scientific study of language (Rossi-Landi 1983, 43).

Bloomfield, alongside Thảo and Rossi-Landi, thought that language denotes material productive activities and forces of production: "In the ideal case, within a group of people who speak to each other, each person has at his disposal the strength and skill of every person in the group." (Bloomfield 1961, 24). For the referential function of language to operate effectively; however, linguistic uniformity is deemed essential. Bloomfield argued that the social division of labor relies on linguistic homogeneity or, at minimum, is significantly enabled by it.

> Obviously, the value of language depends upon people's using it in the same way. Every member of the social group must upon suitable occasion utter the proper speech-sounds and, when he hears another utter these speech-sounds, must make the proper response. He must speak intelligibly and must understand what others say (Bloomfield 1961, 29).

From this perspective, linguistic diversity contrary to the diversity of skills in a social group is perceived as an obstacle to economic cooperation, effectively excluding translation from the processes of labor organization. Translation must be adopted in place of what Bloomfield termed "lines of weakness" in the density of communication (Bloomfield 1961, 47). Lines of weakness are those cleavages which interfere with shared references of the speech community and the referential function of language, and therefore obstruct economic cooperation.

As sociolinguist Ingrid Piller has made clear, "linguistic diversity has long been perceived as an obstacle to development" (Piller 2016, 165). In the middle of the twentieth century, during a phase of intense decolonization, developmental policies have typically associated the former colonial languages with national monolingualism. While statistical analyses since the 1970s have often been used to prove a negative correlation between the number of sizable language

groups and a country's gross domestic product that favors the idea that the greater a country's linguistic diversity, the greater its poverty. Yet, there is no causal relationship between the two terms of this negative correlation, nor statistical evidence that linguistic assimilation would produce the desired outcomes in terms of GDP.

Linguistic anthropologist Judith Temkin Irvine has criticized Bloomfield's univocal reliance on language's propositionality, at the expense of a variety of functions, to assess the effectiveness of the division of labor.

> [D]oes linguistic diversity impede social cooperation? Does the variety of verbal behaviors merely index social groups, divisions, or roles formed on mainly nonverbal bases, or is the variety of verbal performance a precondition for (and thus a defining characteristic of) the social division of labor itself-as the practices constituting a social role, or as the objects of economic activity? (Irvine 1989).

Irvine discusses language's role in political economy by identifying four functions through which verbal signs participate in economic exchange: *propositionality* (signs denote objects and activities in the material world); *indexicality* (signs index social groups and relations of production); *incorporation* (signs function as objects within economic activities); and *authentication* (signs, when pronounced by experts, attest to and confer value upon commodities) (Irvine 1989).

Within Irvine's classification, we can locate several theories that have attempted to read language across the human–machine divide. *Propositionality*, for example, underpins what is today known as the symbol grounding problem (Bender and Koller 2020) or the vector grounding problem (Mollo and Millière 2025). The vector grounding problem refers to the fact that word embeddings are trained on patterns in text alone and are not grounded in sensory, perceptual, or experiential reality. This leads to a disconnect between language representations and the real-world concepts they are intended to denote. We could say, then, that the vector grounding problem is, in fact, a problem of language propositionality. At the other end of the spectrum, I align the *incorporation function* not only with Rossi-Landi's homology of language and labor but also with theories of the "linguistic turn of capitalism" (Marazzi 1994; Virno 2004), which describe how, through "linguistic machines," communication directly enters the production process. In this framework, language and speech do not merely denote or index economic activity; rather, they *constitute* economic activity.

We can interpret the multiplicity of linguistic functions discussed by Irvine in the context of the social division of labor as one example of what Sandro Mezzadra and Brett Neilson describe with the concept of the "multiplication of labor" (Mezzadra and Neilson 2013, 22). This concept, in contrast to Bloomfield's analysis, portrays the division of labor as multiplying alongside the proliferation of borders, both physical and cognitive, reflecting the heterogeneity of workers' statuses, skills, tasks, and languages, which are articulated into hierarchies crucial to the circulation of capital. In this articulation of linguistic and social diversity driven by capital, translation plays a central role. We could argue that it is through translation, understood as the articulation of differences into a hierarchical division of labor that capital operates.

6 Tokenization of labor

Today, the *token* can be considered a critical unit in the NLP pipeline, a metric not unlike Morse code during the Second Industrial Revolution and its colonial expansion, designed to quantify value across different forms of labor. Drawing a parallel with Morse code underscores the broader socio-economic implications of tokenization as a metric applied to language and labor. Yet while highlighting the continuity between the two systems, we should take care not to obscure the specificities and contingencies of their respective applications. Indeed, although both measurement techniques emerged from pragmatic concerns with language compression this applies to Morse code and Byte Pair Encoding tokenization,[7] while WordPiece[8] is more directly optimized for modeling efficiency their purposes diverge in significant ways.

Morse code was a closed system developed to optimize the transmission of messages across constrained technical infrastructures. Tokenization, by contrast, serves a broader computational agenda: it is an open machine-optimized system, which enables the segmentation of language so that it can be processed, quantified, and modeled. In doing so, it lays the groundwork for a new definition of linguistic abstraction, where language is rendered into discrete,

[7] Byte Pair Encoding (BPE) (Sennrich et al. 2016) is a popular tokenization algorithm used in LLMs of the GPT series. First introduced as a data compression algorithm (Gage 1994), BPE allows for variable segmentations based on the internal frequency of each input language, i.e., if two characters often appear together, they will be merged into a subword token.

[8] WordPiece (Schuster and Nakajima 2012) is another popular tokenizer today used by several versions of the model BERT. It is a probabilistic subword modeling technique that was first introduced to obviate the problem of units for languages like Japanese and Korean which do not share the Latin alphabet concept of word boundaries. In Korean spaces occur between particles or grammatical units and not words, while Japanese make use of no spaces but different scripts to identify word boundaries. While not a compression algorithm technically speaking, WordPiece reduces the complexity of language modeling by encoding text in a more efficient and generalizable form.

measurable units that can be operationalized within workflows of machine learning.

Notably, Morse code with its fixed codification of signs, stripping diacritics and script-specific orthographic distinctions offered a deterministic view of linguistic difference, anchored in the English alphabet as its normative center for standardization. In contrast, the language discretization introduced by tokenizers reflects a more complex mediation, one that balances each language's internal statistical coherence with its external dependencies. Contrary to Morse code, tokenization is a form of lossless encoding (Delétang et al. 2023), meaning that the process of turning text into tokens reduces the size of complexity without compromising on the information it carries.

Despite the fact that both measurement techniques emerged from pragmatic concerns with language representation efficiency that did not have immediate fulfillment in the socio-economic domain, and that tokenization can be used in academic and non-commercial contexts it must be acknowledged that technical infrastructures are never neutral and reflect asymmetries across languages in training and development. As discussed earlier in this contribution, tokenization in multilingual LLMs is often biased toward the pretraining data most of which is in English and this bias affects downstream performance. It frequently results in longer token sequences, approaching character-level granularity for underrepresented languages, particularly those that do not use the Latin alphabet. Another key source of tokenization disparity stems from structural and script differences across languages. Many non-Latin scripts, such as Chinese logograms, Korean syllabic blocks, and Indic abugidas do not align neatly with the subword units used in tokenizers trained predominantly on English. Additionally, many tokenizers apply pre-tokenization rules, such as splitting on whitespace or punctuation that reflect alphabet-centric assumptions. These rules can fragment words unnaturally in languages with different grammatical or script conventions, such as Hindi, Tamil, or Japanese.

These asymmetries by which some languages are considered more efficient or more costly than others stem from the system's underlying rationalities. While Morse code was embedded in an industrial logic of efficiency, grounded in standardization as a means of managing time scarcity and increasing throughput, tokenization manifests a statistical rationality that emerges from frequency patterns rather than fixed symbols. Its predictive efficiency is driven by generalization and scale. Both rationalities are embedded in the hierarchies of power that characterize the capitalist impulse toward global communication and the resulting reorganization of the division of linguistic labor.

In this study, I've argued that the problem of linguistic value emerges in concurrence with the social division of labor. Such as the case, for example, with the system of numeralization which allowed characters retrieval as sanctioned by the "alphabetic labor time" of telegraphy, or the translation of value which tokenizers apply to discriminate against the heterogeneity of scripts. How does the automation of translation in multilingual LLMs become part of the articulation of the division of labor? To answer this question, despite a certain anthropocentrism that today would certainly need to be reconsidered, returning to Thåo and Rossi-Landi's proposal of a language emerging from the division of labor can help us to put forward a few hypotheses.

First, investigating the interdependence of language and labor allows us to conceptualize today's "language machines," such as ChatGPT or Google Translate, as genuine machines of collective labor not just of linguistic labor, but encompassing the entire logistics service chain, down to the exploitation of energy resources necessary to power data centers. In fact, for Rossi-Landi, automation arises precisely when material production and linguistic production appear fully reunited, revealing their homology to the point of resolving it into an identity: "[J]ust as soon as we set foot on the ground of material automation, we find ourselves already on the ground of linguistic production as well" (Rossi-Landi 1974, 1884). In this sense, the cost of linguistic labor is also the cost of labor *tout court*.

Secondly, the language–labor nexus prompts us to consider how the discretization introduced by tokenization in language also operates on the plane of labor. Contemporary machine learning models segment tasks according to statistical patterns that are not necessarily grounded in the boundaries of the established division of labor but are instead based on what optimizes predictive accuracy under computational constraints. Digital platforms such as Uber, Airbnb, and Netflix exemplify how previously continuous practices transport, accommodation, and entertainment are rearticulated as discrete, distributed, and modelable units. Whereas the mechanical discretization characteristic of industrial labor imposed rigid and standardized boundaries, contemporary tokenization enacts a probabilistic grammar of segmentation that reorganizes practices according to algorithmic logics of prediction and optimization.

Uber's H3 Hexagonal Hierarchical Spatial Index (Brodsky 2018), for instance, partitions the earth's surface into a hierarchical grid of hexagons, producing a modularized spatial order within which demand forecasting, route optimization, and dynamic ride pricing become computationally actionable. The driver's labor, in turn, is reorganized around these probabilistic trajectories. Each ride is broken down into modular stages: the assignment of a request (ping), the accept/decline decision within seconds, the algorithmically optimized route to the pickup point, the ride itself, and the final rating exchange. For researchers in the field of spatial data modeling, "trajectories are like statements" (Musleh et al. 2022), with each word or token corresponding to a

GPS point and the road network and physical constraints functioning as their grammar. In this context, "spatial tokenization" can be designed to process raw data, such as anonymized user IDs, coordinates, and timestamps, and infer motion sequences (Najjar 2023; Xu et al. 2025).

Asserting that "trajectories are like statements" might seem a fitting example of the projection of linguistic frameworks onto other disciplines, an approach already critiqued as the deployment of a "meta-science" of general pattern machines (Offert et al. 2026). Yet Rossi-Landi invites us to think in terms of a structural homology between language and labor, rather than asserting the primacy of linguistics. It is important to acknowledge that LLMs introduce an alternative grammar of segmentation one that demands new conceptual tools for understanding how tokenization interacts with historically grounded divisions of labor.[9] These interactions range across domains from computational linguistics to trajectory modeling, protein folding, and blockchain technologies applied to agricultural supply chains. Addressing the question of the cost of language thus requires us to investigate the token both as a technical and a socio-economic measure, considering its implications not only for the efficiency of linguistic machines but also for the broader socio-economic structures they reinforce or disrupt, and its intersections with the global division of labor.

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[9] When Leif Weatherby describes "language as a service" as an essential feature of platform capitalism and distinguishes the "labor of writing" from the "management of writing" (Weatherby 2025, 196), he captures an important dimension of the automation of language in relation to the division of labor. LLMs, however, do not merely automate the "labor of writing", the tedious, low-skill tasks that require no specific expertise, such as email composition, accounting, or proofreading (though we should not underestimate the relational intelligence required to carry out even these mundane operations). Since Charles Babbage, the principle of automation has involved not only the imitation and replacement of the existing division of labor through the mechanization of lower-level tasks, but also the more precise measurement of the cost of each task the so-called "Babbage principle" of labor calculation (Pasquinelli 2023, 63). In this sense, the design of the machine, by automating and measuring textual production, provides both language as a service and its management. A vivid illustration of this dual nature of automation can be found in editing tools such as Rubriq, which are embedded in contemporary academic publishing platforms and automatically evaluate the linguistic quality of a text while simultaneously offering technical "services" to improve its accuracy.

Declarations

Conflict of interest The authors declare no competing interests.

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